

Euclid’s Elements — Book I

Proposition 46

Formal Reconstruction — Technical Documentation

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Constructing a square on a given segment

1 Introduction

Proposition I.46 constructs a square on a prescribed segment. It is the last construction immediately before Proposition I.47, and in the present formalization it supplies exactly the square data needed by the Pythagorean theorem.

The proposition is geometrically much cleaner than the equal-content block I.35–I.45. No scissors relation, equicomplementability relation, polygon representation, positivity principle, or numerical area is involved. The proof returns to incidence, order, segment congruence, right angles, parallels, and the developed Euclidean theory of parallelograms.

The final Lean proof has two levels.

- A proposition-local helper, `i46_erect_equal_perpendicular`, constructs the second side of the square at the endpoint of the given segment.
- The main theorem completes the parallelogram, obtains the four equal sides from I.34, and obtains the four right angles from the newly packaged parallelogram-angle interface.

A significant architectural result emerged during the formalization. The initial proof of the adjacent-right-angle step contained a long SameSide/OppositeSide and transversal normalization argument. That argument was not specific to squares. It was factored into reusable parallelogram lemmas in `HilbertInterface`. Proposition I.46 now consumes the general theorem

```
parallelogram_adjacent_right_angle
```

instead of carrying the I.29 normalization locally.

The final source contains no proposition-local geometric axiom and no `sorry`.

2 The Classical Configuration

Let AB be the given segment. The final square is written in the Lean vertex order

$$A, B, C, D.$$

Thus AB is the given side, AD is the erected perpendicular side, and the constructed quadrilateral is

$$ABCD.$$

The classical construction may be read as follows.

1. At A , erect a perpendicular to AB .
2. On that perpendicular choose D so that

$$AD \cong AB.$$

3. Through D draw the parallel to AB , and through B draw the parallel to AD . Their intersection is C .
4. The quadrilateral $ABCD$ is then a parallelogram.
5. Proposition I.34 makes the opposite sides congruent; since $AD \cong AB$, all four sides are congruent.
6. Proposition I.29 transfers the right-angle information to the adjacent vertex, and I.34 transfers rightness to the opposite angles.

Figure 1 records only this classical construction. The formal witnesses needed to implement the endpoint perpendicular are kept out of the classical diagram and are displayed separately in Figure 2 below.

3 Classical Proof

Euclid's proof uses the standard Book I construction chain.

First apply I.11 to erect a perpendicular at the endpoint A of the given segment AB . Then use I.3 to cut off on that perpendicular a segment AD equal to AB . Through D draw the parallel to AB , and through B draw the parallel to AD ; this is the I.31 parallel construction.

The resulting quadrilateral is a parallelogram. By I.34 its opposite sides are equal:

$$AB \cong CD, \quad BC \cong DA.$$

Since construction gave

$$DA \cong AB,$$

transitivity yields equality of all four sides.

It remains to show that all four angles are right. The given angle at A is right by construction. Since one side of the parallelogram is a transversal of the opposite parallel sides, I.29 gives the appropriate supplementary or alternate-angle relation at the adjacent vertex. Because the angle

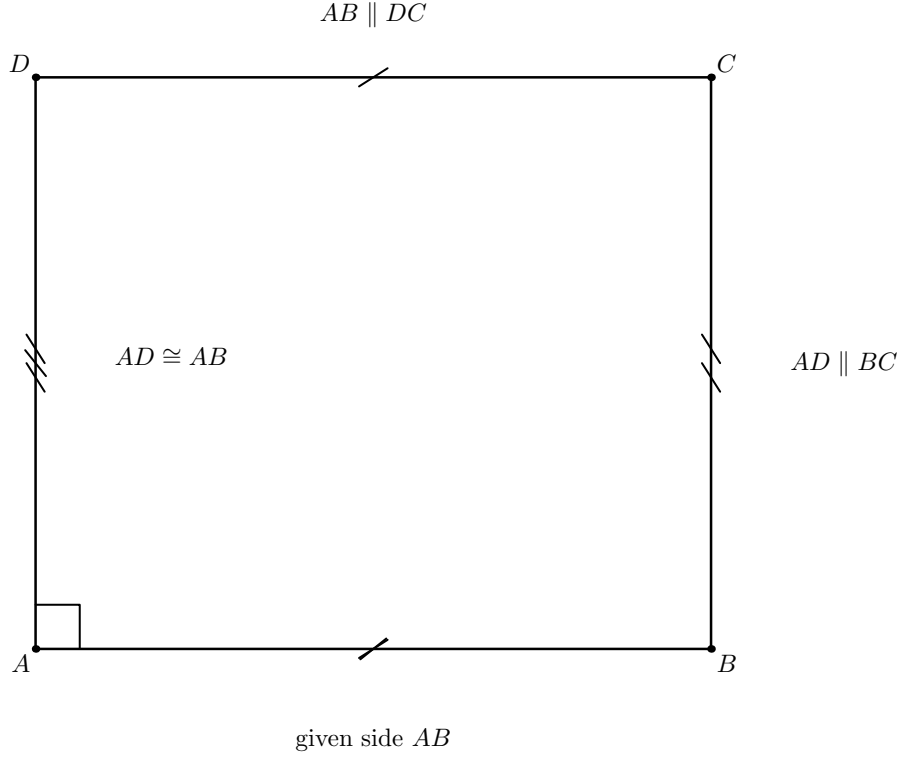


Figure 1: The classical I.46 configuration. The prescribed side is AB ; at A one erects the equal perpendicular side AD , and the parallels through D and B meet at C . Thus $ABCD$ is the parallelogram from which I.34 and I.29 complete the proof that all four sides are equal and all four angles are right. No formal endpoint witnesses are shown here.

at A is right, the adjacent interior angle is also right. Proposition I.34 then gives the congruence of opposite angles, so the two remaining angles are right.

The historical dependency may therefore be summarized as

$$\begin{aligned}
 &AB \text{ given} \\
 &\Downarrow \text{ I.11, I.3} \\
 &AD \perp AB, \quad AD \cong AB \\
 &\Downarrow \text{ I.31} \\
 &ABCD \text{ parallelogram} \\
 &\Downarrow \text{ I.34} \\
 &AB \cong BC \cong CD \cong DA \\
 &\Downarrow \text{ I.29, I.34} \\
 &\text{all four angles right} \\
 &\Downarrow \\
 &ABCD \text{ square.}
 \end{aligned}$$

4 The Square Predicate and the Representation Choice

Euclid defines a square as an equilateral and right-angled quadrilateral. The Lean development records slightly more structure: the four vertices must also form a parallelogram.

In documentation-friendly ASCII notation, the definition is

```
def IsSquare
```

```

(A B C D : Geo.Point) : Prop :=
IsParallelogram Geo A B C D /\
Geo.Congruent A B B C /\
Geo.Congruent B C C D /\
Geo.Congruent C D D A /\
HilbertRightAngle Geo D A B /\
HilbertRightAngle Geo A B C /\
HilbertRightAngle Geo B C D /\
HilbertRightAngle Geo C D A

```

This is a representation strengthening, not an additional geometric claim proved later. The construction already produces a parallelogram, so the extra clause records the incidence structure that distinguishes the intended quadrilateral from a crossed four-point configuration.

For Proposition I.46 this choice is useful in two ways.

- Opposite-side and opposite-angle theorems can be applied directly.
- Downstream propositions can consume a square together with its parallelogram structure without reconstructing the boundary order from the four metric and angular clauses.

The representation remains entirely synthetic. No coordinates, side lengths as numbers, or area values are introduced.

5 Proof Architecture of the Reconstruction

The current Lean proof is best read as five conceptual stages.

5.1 Step 1: erect an equal perpendicular side

The local theorem

```
i46_erect_equal_perpendicular
```

takes distinct points A, B and constructs a point D such that

$$A, B, D \text{ are noncollinear,} \quad \angle DAB \text{ is right,} \quad AD \cong AB.$$

The available right-angle constructor is formulated at an interior point of a straight line. Therefore the proof first extends BA beyond A :

$$B - A - F.$$

It then applies the nondegenerate right-angle existence theorem to obtain a point X off the carrier BA . Hilbert III.1, represented by `segment_construction`, lays off the length AB on the ray AX , producing D .

The last part of the helper is representational. Since D and X lie on the same ray from A , the angle with arm AX is transported to the angle with arm AD . The unoriented angle calculus then exchanges the arms so that the output has exactly the orientation required by the main theorem:

$$\text{HilbertRightAngle}(D, A, B).$$

Figure 2 separates the two formal states of this helper. This separation is important: the proof establishes that D lies on the same ray from A as X , but it does not establish an order relation such as $A - X - D$ or $A - D - X$. The first panel therefore shows the direction point X , while the second panel shows the laid-off point D on the chosen ray without placing X and D simultaneously.

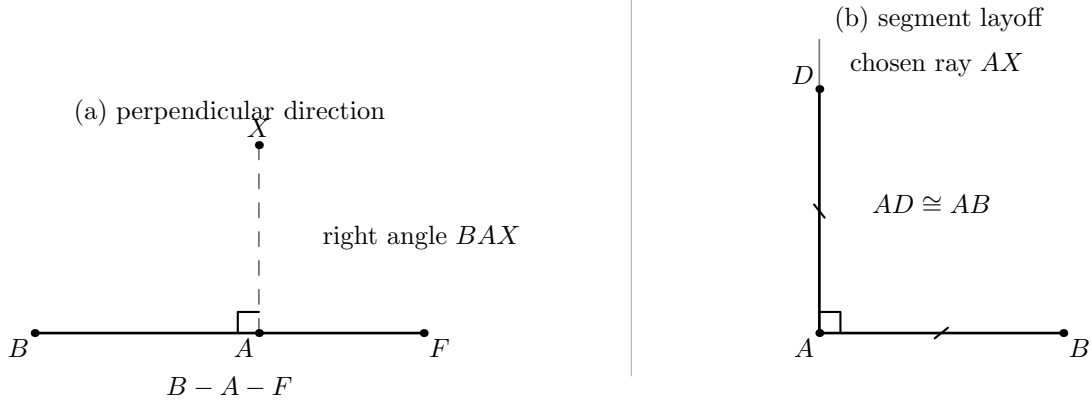


Figure 2: The formal implementation of the endpoint construction. In (a) the segment BA is extended to $B - A - F$, so the developed I.11 theorem can construct a nondegenerate right angle BAX . In (b) Hilbert III.1 lays off the length AB on that chosen ray, producing D with $AD \cong AB$. The panels deliberately do not impose any betweenness relation between X and D ; the Lean proof uses only the `SameRay` relation.

5.2 Step 2: complete the parallelogram

From noncollinearity of A, B, D , the theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

constructs the fourth point C and returns

$$\text{IsParallelogram}(A, B, C, D).$$

This theorem packages the two parallel constructions and their intersection. The public I.46 proof does not replay carrier uniqueness or parallel transitivity at this point.

5.3 Step 3: prove all four sides congruent

The main theorem invokes Proposition I.34:

```
euclid_proposition_34
```

and extracts

```
OppositeSidesCongruent Geo A B C D
```

Together with the constructed equality

$$AD \cong AB,$$

ordinary congruence symmetry and transitivity produce

$$AB \cong BC, \quad BC \cong CD, \quad CD \cong DA.$$

No new segment theorem is introduced here. The proof is a short calculation in the established congruence API.

5.4 Step 4: propagate the right angle to the adjacent vertex

This was the main refactoring point of I.46.

The final proposition now calls

```
parallelogram_adjacent_right_angle
```

with the parallelogram $ABCD$ and the right angle DAB . The result is

$$\angle ABC \text{ right.}$$

The theorem is backed by the more geometric interface lemma

```
parallelogram_adjacent_exterior_angle_congruent
```

which extends the side CB through B to a point F :

$$F - B - C,$$

and proves

$$\angle DAB \cong \angle FBA.$$

Its proof is the exact I.29 configuration-normalization that was initially local to Proposition I.46. The line AB is fixed as a transversal. The parallel sides of the parallelogram provide SameSide data for C, D . Extending CB produces F on the opposite side of the transversal. The parallel relation is transported from BC to BF , and Hilbert's Euclidean alternate-angle theorem gives the required exterior-angle congruence.

Figure 3 displays this reusable configuration.

5.5 Step 5: obtain the two opposite right angles

I.34 also returns

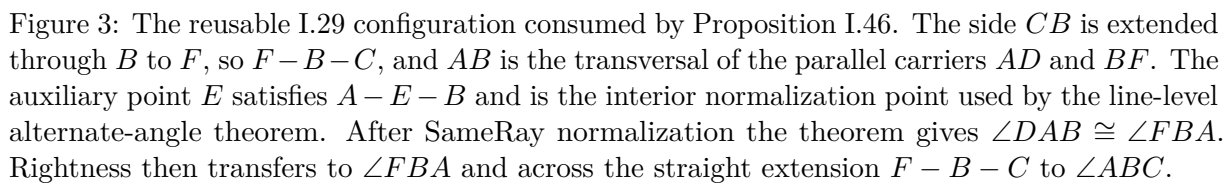
```
OppositeAnglesCongruent Geo A B C D
```

The established right angle DAB is transported to the opposite angle BCD . The newly established right angle ABC is transported to the opposite angle CDA . The helper

```
parallelogram_vertices_noncollinear
```

supplies all four nondegeneracy conditions required by the angle transport theorems.

The eight clauses of `IsSquare` are then assembled directly.



The current proposition imports exactly the two earlier Euclidean propositions used explicitly by the source:

Earlier parallel and Book Zero infrastructure enters transitively through those modules and through `HilbertInterface`.

In ASCII notation, the proposition-local helper has the signature

7

```

exists D : Geo.Point,
  Not (Collinear Geo A B D) /\
  HilbertRightAngle Geo D A B /\
  Geo.Congruent A D A B

```

This theorem is local because its packaging is tailored to the exact input required by the square construction: an equal perpendicular side based at the first endpoint of the given segment.

6.2 The public proposition

The public theorem is

```

theorem euclid_proposition_46
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B : Geo.Point)
  (hAB : Ne A B) :
  exists C D : Geo.Point,
    IsSquare Geo A B C D

```

The proof first constructs D , then constructs C , obtains I.34 once, and uses the resulting side and angle packages twice. After the interface refactor, no SameSide/OppositeSide argument remains in the public proposition.

6.3 The downstream side package

For Proposition I.47 the file exposes

```

theorem euclid_proposition_46_side
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hSquare : IsSquare Geo A B C D) :
  Geo.Congruent A B B C /\
  Geo.Congruent A B C D /\
  Geo.Congruent A B D A

```

This theorem does not add geometry. It packages the three side congruences in the orientation used by the Pythagorean construction.

6.4 Reusable helpers promoted to the interface

The formalization of I.46 produced four reusable declarations:

```

parallel_first_not_collinear
parallelogram_vertices_noncollinear
parallelogram_adjacent_exterior_angle_congruent
parallelogram_adjacent_right_angle

```

The first two are neutral consequences of strict parallelism and order. The last two are Euclidean:

they use the direction

parallel lines $\implies$ congruent alternate angles,

which is the Hilbert Group IV side of Theorem 30.

This extraction is an architectural improvement rather than a change in the mathematics of I.46. The proposition now consumes a general parallelogram theory in the same way that I.34 and later propositions consume the previously developed opposite-side and opposite-angle API.

7 Relationship with Earlier Propositions

7.1 Proposition I.11

The first local helper is the endpoint form of the perpendicular construction. The project theorem used internally is

```
hilbert_right_angle_exists_nondegenerate
```

which corresponds to the developed Book Zero/Hilbert form of I.11.

Because its hypotheses use a point strictly between two collinear points, the proof first produces BA beyond A . This is hidden in the ordinary Euclidean instruction “draw a perpendicular at A ” but explicit in the Lean data.

7.2 Proposition I.3 and Hilbert III.1

Euclid uses I.3 to cut off

$$AD = AB.$$

The reconstruction performs the same geometric operation through Hilbert’s segment-construction axiom/theorem:

```
HilbertCongruence.segment_construction
```

The point D is not obtained merely as an arbitrary point on the perpendicular line. It is constructed on the chosen ray, which preserves the orientation needed by the subsequent angle transport.

7.3 Proposition I.29

I.29 is the unique visibly Euclidean angle step of the proposition. In the interface it appears through the theorem converting parallelism into alternate angle congruence.

The important formal point is that the final I.46 file no longer performs the transversal normalization itself. That work has been generalized to the two interface theorems

```
parallelogram_adjacent_exterior_angle_congruent  
parallelogram_adjacent_right_angle
```

7.4 Proposition I.31

The classical instruction “draw parallels through D and B ” is consumed through

```
hilbert_parallelogram_fourth_vertex_exists
```

This theorem packages the existence of the fourth vertex and returns the `IsParallelogram` structure directly.

7.5 Proposition I.34

I.34 is the principal direct proposition dependency. One call returns both the opposite-side and opposite-angle data. I.46 uses the first component to make the quadrilateral equilateral and the second component to transport the two already known right angles to the opposite vertices.

7.6 Proposition I.47

The construction is designed for immediate reuse by I.47. The helper

```
euclid_proposition_46_side
```

exposes the side congruences of a square in a convenient orientation. No additional square theory needs to be reconstructed in the next proposition.

8 Hilbert and Book Zero Components Used

8.1 Betweenness and extension

The endpoint perpendicular construction requires an explicit produced segment

$$B - A - F.$$

The interface lemma `ExtendSegmentBeyond` supplies this witness. The strict order is not decorative: it is what makes the nondegenerate right-angle constructor applicable at A .

The adjacent-angle theorem uses another produced segment

$$F - B - C.$$

That strict relation is what allows a right exterior angle at B to be transported across the straight line to the interior angle ABC .

8.2 SameRay and angle invariance

After the perpendicular direction AX has been constructed, Hilbert III.1 places D on the same ray from A . The equality

$$\angle BAX = \angle BAD$$

is therefore a ray-invariance statement, not a metric calculation.

The proof then uses the project’s unoriented angle calculus to reverse the two arms and obtain the orientation DAB expected by the square predicate.

8.3 Parallel carriers and plane separation

The extracted exterior-angle theorem uses the developed carrier-line theory. Its proof fixes the line AB as a transversal, places the relevant endpoints on the required sides of that line, extends the second parallel carrier, and transports the parallel relation to the produced point.

The public I.46 theorem does not expose these witnesses, but they are genuine logical data behind the convenient theorem `parallelogram_adjacent_right_angle`.

8.4 Right-angle transport

Rightness is represented synthetically by `HilbertRightAngle`; no numerical angle measure is present.

The theorem

```
hilbert_right_angle_transport
```

is used repeatedly. Once two nondegenerate angles are congruent, rightness of one angle can be transported to the other. This is how I.34's opposite-angle congruences become right-angle statements.

9 Formalization Notes

9.1 Geometry

The hidden configuration data are concentrated in two places.

For the local endpoint construction Lean must record:

- $B - A - F$ after extending the given carrier;
- noncollinearity of the perpendicular point;
- the `SameRay` relation placing D on the perpendicular ray;
- the noncollinearity permutations required by angle transport.

For the reusable adjacent-angle theorem the interface proof records:

- the carrier of AB ;
- `SameSide` data for the two relevant parallelogram vertices;
- the extension $F - B - C$;
- `OppositeSide` data after the extension;
- transport of the parallel relation from BC to BF ;
- an interior point on the transversal used by the line-level form of Hilbert Theorem 30;
- `SameRay` normalization of the resulting angle arms.

These facts are exactly the kind of information suppressed by the classical diagram but required by a formal synthetic proof.

9.2 Content

There is no content layer in Proposition I.46.

This is important because I.46 follows immediately after the long area and equicomplementability development I.35–I.45. The proposition uses none of the following:

```
HilbertScissorsEq
HilbertScissorsEquicomplementable
equicomplementable_add
content comparison
proper-part positivity
```

No cancellation, no de Zolt principle, no multiplication of segment lengths, and no numerical area is involved. The proof is ordinary Euclidean construction geometry.

9.3 Representation

The main representation decision is the definition

```
IsSquare
```

whose stored data include an explicit

```
IsParallelogram
```

clause in addition to the four equal-side and four right-angle conditions.

This is stronger as stored data than Euclid’s verbal Definition I.22, but it does not strengthen the constructed conclusion beyond the actual geometry: the proof has already built the figure as a parallelogram. The clause simply retains useful incidence and parallel structure for downstream use.

9.4 Axiomatic status

Proposition I.46 is Euclidean.

Hilbert Group IV enters in the developed theory of parallels. In particular, the theorem producing the adjacent-angle congruence uses the Euclidean direction of Hilbert Theorem 30:

parallel $\implies$ equal alternate angles.

The fourth-vertex construction also relies on the developed Euclidean parallel infrastructure needed to identify the appropriate intersection.

The endpoint perpendicular and segment-layoff construction is neutral: it uses incidence, order, and congruence, but not the uniqueness of parallels.

No continuity axiom is used. No Archimedean axiom is used. No new geometric axiom is introduced. No content axiom is used.

10 Source Audit

The source audit separates the historical proof from the formal architecture.

Euclid, together with the standard Heath/Joyce and Wilkins presentations, controls the classical dependency chain: erect the perpendicular, cut off the equal side, draw the two parallels, use I.34 for opposite sides, and use I.29 plus I.34 for the right angles.

Hilbert controls the logical placement of the parallel-angle step. The direction from equal alternate angles to parallel lines belongs to the neutral development, while the converse direction from parallel lines to equal alternate angles uses Group IV. Proposition I.46 therefore crosses the Euclidean boundary precisely where the ordinary construction expects it.

Borsuk–Szmielew and Greenberg are control sources for the developed theory of parallel carriers, parallelograms, and plane separation. They support placing the adjacent-angle result in the reusable Euclidean interface rather than leaving it as a proposition-local fact.

The Beeson proof corpus is useful here mainly as a check on hidden configuration conditions: endpoint extension, ray orientation, explicit transversal data, and noncollinearity. The final proof is not organized around Tarski-style tactics.

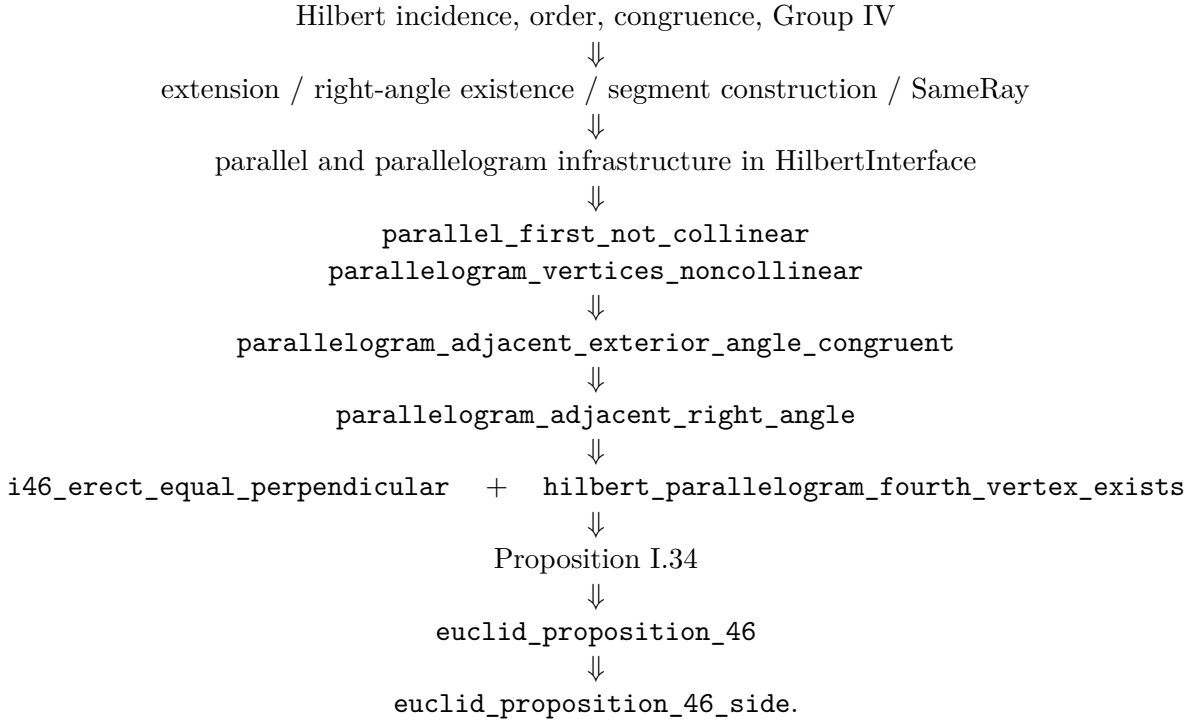
Hartshorne adds no special content issue at I.46. In contrast with I.35–I.45, there is no equal-content semantic boundary to control.

The three-way audit requested for propositions from I.35 onward is therefore particularly simple:

- **Geometry:** nontrivial; perpendicular construction, segment layoff, parallel completion, I.29, and I.34.
- **Content:** absent.
- **Representation:** the explicit `IsParallelogram` clause in `IsSquare`, plus the packaging theorem prepared for I.47.

11 Dependency Summary

The formal dependency chain is



The classical and Lean DAGs can be compared succinctly as

Euclid: I.11 + I.3 $\rightarrow$ I.31 $\rightarrow$ I.34 $\rightarrow$ I.29/I.34,
 Lean: endpoint extension + segment construction $\rightarrow$ fourth-vertex theorem
 $\rightarrow$ I.34 $\rightarrow$ interface adjacent-right-angle theorem.

New geometric axiom:	no
New content axiom:	no
New proposition-local helper:	yes
New reusable helper:	yes
New Book Zero theorem:	no
New Interface theorem:	yes, four
Numerical area semantics:	no
Continuity used:	no
Archimedean axiom used:	no
Hilbert Group IV used:	yes
Public theorem compiles:	yes
Full lake build:	clean
Active local axiom:	no
sorry in Proposition46.lean:	no

Diagrammatic audit. The final figure set separates the three geometric roles that were conflated in the previous presentation. Figure 1 is the classical Euclidean construction and therefore contains no proposition-local formal witnesses. Figure 2 records the two successive states of `i46_erect_equal_perpendicular`; in particular it does not turn the `SameRay` conclusion for X and D into an unsupported betweenness claim. Figure 3 records the genuinely geometric configuration hidden behind the reusable adjacent-right-angle interface, including the

normalization point E with $A - E - B$. No additional figure is introduced for the fourth-vertex theorem, the I.34 congruence bookkeeping, or `euclid_proposition_46_side`, since those steps do not create a new geometric state needed by the proposition-level argument.

12 Conclusion

Proposition I.46 is a compact construction theorem whose formalization exposes two useful structural facts.

First, the endpoint version of the perpendicular construction is not definitionally trivial in the Hilbert API. It requires an extension witness, a nondegenerate perpendicular direction, a segment layoff on a chosen ray, and angle transport. These details are packaged locally in `i46_erect_equal_perpendicular`.

Second, the ordinary statement that an adjacent angle of a parallelogram is right when one angle is right is not a square-specific fact. Its proof is a general application of I.29 together with plane-separation and ray normalization. Extracting that argument into `HilbertInterface` removes a long low-level configuration proof from I.46 and strengthens the reusable Euclidean parallelogram API.

After that extraction the public proposition has the expected classical shape: construct one equal perpendicular side, complete the parallelogram, use I.34 for opposite sides and angles, and assemble the square. The resulting `IsSquare` object is immediately prepared for Proposition I.47 through `euclid_proposition_46_side`.